

C. Balkı GEMİRTER

Chief AI Officer (CAIO) & R&D Leadership

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Location İstanbul, Türkiye **Languages** Turkish (Native) | English (Professional) | Japanese (Elementary)

Artificial Intelligence & Deep Learning | HealthTech Solutions | FinTech & Predictive Analytics | R&D & Innovation Management | Large Language Models (LLM) & NLP | IoT & Smart City Analytics

19+ YEARS OF EXPERIENCE | **6** PUBLICATIONS | **9** R&D PROJECTS | **16** CERTIFICATES

PROFILE

Ph.D. candidate in Computer Engineering at Yeditepe University (4.00 GPA) with 19+ years of enterprise R&D leadership and deep-tech entrepreneurship experience. Built upon a strong corporate engineering foundation at TEB and Nortel/Netaş, with concurrent leadership in 0-to-1 AI architectures, TÜBİTAK-funded projects, and LLM implementations as CAIO and AI Director across high-growth startups (CEIBA Health, Bayegan, Agada Tech, Vivoo). Specialized in transforming advanced AI research into scalable products and measurable business value in healthcare, finance, and commodity trading.

EXPERIENCE

STARTUP & TRANSFORMATION LEADERSHIP — 2021 – Present

AI strategy and product architecture build-outs at deep-tech startups, carried out concurrently with doctoral research and university teaching.

Yeditepe University — Sep 2021 - Present
Ph.D. Candidate · Teaching Assistant & Guest Lecturer
- Doctoral research on incorporating prior knowledge into deep learning architectures for image recognition and medical diagnosis with small cohort data (expected graduation: December 2026)
- Teaching assistant for Artificial Intelligence (CSE 462), Machine Learning (CSE 585) and Deep Learning (CSE 588)
- Designs and grades the term projects and assignments of these courses
- Guest lecturer for the university-wide AI 401 course and at HSE University (Moscow)

CEIBA HEALTH — Jan 2026 - Apr 2026
Director of Artificial Intelligence
- Led AI and R&D strategy for next-generation healthcare solutions
- Secured and managed TÜBİTAK 1501 and 1832 projects
- Directed YOLO-based real-time fall risk detection (>90% accuracy, <10s alert latency) and architected both the LLM-powered "AI Companion" for automated SOAP notes and a DSP-based strip analysis system for ECG rhythm analysis

Bayegan — May 2025 - Jul 2025
Chief AI Officer (CAIO)
- Spearheaded the AI roadmap of a commodity trading company, aligning it with business objectives to enhance predictive analytics and risk management
- Directed model training and optimization for automated trading systems
- Fostered collaboration with trading desks and software engineers to integrate AI into workflows

Agada Tech — Sep 2024 - Jan 2025
VP of Artificial Intelligence
- Spearheaded AI-driven customer segmentation models and oversaw real-time next-order prediction systems for high-volume e-commerce
- Pioneered advanced AI methodologies to enhance customer loyalty strategies

Yapı Kredi Teknoloji — May 2024 - Aug 2024
AI & Applied Data Science Manager
- Led a team of 17 engineers with 0% attrition
- Managed critical AI projects such as Digital Mizan and Fraud Detection
- Delivered cost and FTE savings through computer vision, anomaly detection, and deep learning
- Led ITU Technopark-supported R&D and contributions to Horizon 2020

Vivoo — Aug 2023 - Feb 2024
Chief AI Officer (CAIO) & Data Analytics
- Managed the development of "Welly", an LLM-based wellness assistant, and AI-driven test strip analysis tools
- Led a TÜBİTAK 1501-supported project analyzing nutritional values from meal photos

ENTERPRISE R&D & SOFTWARE MANAGEMENT 2007 – 2023

R&D, team leadership, and patent/TÜBİTAK processes within large-scale banking and telecommunications infrastructures.

TEB - Türk Ekonomi Bankası — 4 roles — Jan 2013 - Aug 2023

Senior R&D Manager (Artificial Intelligence, Software) — Aug 2016 - Aug 2023

- Led a team of 10 engineers in TEB's R&D department
- Idea owner and supervisor of the TÜBİTAK TEYDEB 1501-supported NEURON (AI chatbot) and ConnecTEB projects
- Led deep learning/NLP initiatives such as Dragon and Ribbon
- Pioneered the transition of the first R&D team to TEB ARF

Infrastructure Software Development Assistant Project Leader — Oct 2015 - Aug 2016

- Architect of the core IT framework used by TEB's software designers across all banking projects
- Led scalability, performance, and new-requirement integration efforts

Switch and ATM Software Development Assistant Project Leader — Nov 2014 - Oct 2015

- Core member of the Aquila team on TEB's first TÜBİTAK TEYDEB 1501 project, a 100% in-house XTM system
- Led a team of 5 engineers from prototype to full-scale deployment

Internet Banking Software Development Project Supervisor — Jan 2013 - Nov 2014

- Team lead for over 10 engineers across retail and commercial internet banking
- Spearheaded the redesign of both platforms and delivered an advanced smart search feature

NORTEL Networks - NETAŞ — 2 roles — Jul 2007 - Jan 2013

Senior Software Design Engineer — Mar 2010 - Jan 2013

- Designed and developed the A2 Base/OAM system manager and its sub-components using Java and Oracle
- Built high-performance, resilient architectures for critical network infrastructure

Junior Software Design Engineer — Jul 2007 - Mar 2010

- Developed and tested system management (OAM) software components in Java for telecommunications network infrastructure

EDUCATION

Ph.D. - Computer Engineering — 2020 - Dec 2026 (Expected)

Yeditepe University — Taught in English

4.00 / 4.00 — Ranked 1st, Full Scholarship

THESIS

Incorporating prior knowledge in Deep Learning (DL) architectures for image recognition and medical diagnosis with small cohort data.

M.Sc. - Computer Engineering — 2018 - 2020

Yeditepe University — Taught in English

3.87 / 4.00 — Ranked 1st

THESIS

A Turkish Question Answering System Based on Deep Learning Networks (BERT).

B.Sc. - Computer Engineering — 2002 - 2007

Ege University

3.34 / 4.00 — Ranked 4th

Education in core computer engineering disciplines and software architecture.

88 — **ALES** 2018 — Academic Personnel and Postgraduate Education Entrance Exam

85 — **YÖKDİL** 2026 — Higher Education Institutions Foreign Language Exam (English)

ACADEMIC PUBLICATIONS

01 **RP-Fuse: Development of an Adaptive Dose-Gated Multimodal Model for Predicting Radiation Pneumonitis in Non-Small Cell Lung Cancer** SCIE - Q2
MDPI - Applied Sciences (2026) | C. Balkı GEMİRTER, Esra KORKMAZ KIRAKLI, Cemre Ceren ÖZDİLLİ, Ali Kadri ÇIRAK, E. Erkan KORKMAZ, Dionysis GOULARAS

02 **CoDA: A Cognitive-Inspired Approach for Domain Adaptation** SCIE - Q2
MDPI - Applied Sciences (2026) | C. Balkı GEMİRTER, E. Erkan KORKMAZ, Dionysis GOULARAS

03 **Location Proofing Using Video Similarity** TR Dizin
Journal of Scientific Reports-A (2022) | C. Balkı GEMİRTER, Tacha SERIF

04 **A Turkish Question Answering System Based on Deep Learning Neural Networks** TR Dizin
JISTA - Journal of Intelligent Systems (2021) | C. Balkı GEMİRTER, Dionysis GOULARAS

05 **Fine-Tuning Throughput and QoS on SMT Cores** IEEE
UBMK 2021 - 6th International Conference on Computer Science and Engineering (2021) | C. Balkı GEMİRTER, Gürhan KÜÇÜK

06 **A Comparative Evaluation of AMQP, MQTT and HTTP Protocols Using Real-Time Public Smart City Data** IEEE
UBMK 2021 - 6th International Conference on Computer Science and Engineering (2021) | C. Balkı GEMİRTER, Çağatay ŞENTÜRCA, Şebnem BAYDERE

TÜBİTAK & R&D PROJECTS

TÜBİTAK TEYDEB 1501 · 2026

Image Processing-Supported Patient Monitoring and Fall Risk Management

CEIBA HEALTH

Supervised R&D project focusing on AI-based healthcare solutions.

TÜBİTAK TEYDEB 1832 · 2026

Green and Sustainable Digital Solutions in the Health Sector

CEIBA HEALTH

Supervised R&D initiative for eco-friendly digital health.

TÜBİTAK TEYDEB 1501 · 2024 – 2025

AI-Based Dynamic Customer Segmentation and Personalized Gamification System

Agada Tech

Advanced AI models for user retention and personalized experiences.

TÜBİTAK TEYDEB 1501 · 2023 – 2024

AI-Based Nutrition Analysis System

Vivoo

Multimodal health AI project expanding personalized nutrition insights.

Boğaziçi Technopark · 2022 – 2023

Dragon - Banking Question Answering (MRQA) System

TEB - Türk Ekonomi Bankası

Machine reading comprehension-based smart search over banking documents, giving call center staff instant, concise answers to banking queries.

TÜBİTAK TEYDEB 1501 · 2020 – 2022

AI-Based Customer Chatbot

TEB - Türk Ekonomi Bankası

Developed NEURON, an AI chatbot integrated into digital banking.

TÜBİTAK TEYDEB 1501 · 2016 – 2018

ConnecTEB - VoIP & Multimedia Solution

TEB - Türk Ekonomi Bankası

In-house secure video conferencing tool adopted bank-wide.

TÜBİTAK TEYDEB 1501 · 2014 – 2016

Orion & Aquila (In-House XTM)

TEB - Türk Ekonomi Bankası

Core IT framework development and scaling of ATM/Switch infrastructure.

EU-Supported

InVisIBLe Plus

Yeditepe University

Inclusive and innovative learning tool for visually impaired and blind people.

ACADEMIC & PROFESSIONAL SERVICE

Turkish Journal of Electrical Engineering & Computer Sciences

Reviewer · TÜBİTAK

European Journal of Radiology

Reviewer · Elsevier

Bilişim Vadisi (Informatics Valley)

Mentor · Technopark

HSE University

Guest Lecturer · Spring 2025 & 2026

Invited 3-hour Artificial Intelligence lectures at HSE University (Moscow) in two spring terms; designed the course term projects.

Yeditepe University

Teaching Assistant · 2021 – Present

Teaching assistant to Prof. Dr. E. Erkan Korkmaz for Artificial Intelligence (CSE 462, AI 401), Machine Learning (CSE 585) and Deep Learning (CSE 588), and to Dr. Onur Demir for Algorithm Analysis (CSE 311); designed and graded term projects and assignments.

CERTIFICATIONS & AWARDS

AWARD

Leaders of Tomorrow (x2)

Global Talent Award · 2023 - 2024

Scrum.org

PSPO-I Professional Scrum Product Owner

Mar 2020

Scrum.org

PSM-I Professional Scrum Master

Feb 2017

Generative AI for Everyone — Mar 2025
DeepLearning.AI

Sentiment Analysis with Deep Learning using BERT — Feb 2022
Coursera

Deep Neural Networks with PyTorch — Aug 2021
IBM

Natural Language Processing Specialization — Dec 2020
DeepLearning.AI

Machine Learning — May 2020
Stanford Online

TensorFlow in Practice Specialization — Mar 2020
DeepLearning.AI

AI for Medicine Specialization — Jul 2022
DeepLearning.AI

Scalable Machine Learning on Big Data using Apache Spark — Nov 2021
IBM

Generative Adversarial Networks (GANs) Specialization — Feb 2021
DeepLearning.AI

Fundamentals of Parallelism on Intel Architecture — Nov 2020
Intel

Deep Learning — Apr 2020
DeepLearning.AI